Automatica 80 (2017) 54–61

Contents lists available at ScienceDirect

Automatica

journal homepage: www.elsevier.com/locate/automatica

Brief paper

Complete offline tuning of the unscented Kalman filter[☆]

Leonardo Azevedo Scardua [a,b], José Jaime da Cruz [b]

[a] Federal Institute of Espírito Santo, Serra, ES, Brazil

[b] Automation and Control Laboratory, University of São Paulo, São Paulo, SP, Brazil

ARTICLE INFO

Article history:
Received 1 June 2015
Received in revised form
20 November 2016
Accepted 16 January 2017

Keywords:
State estimation
Nonlinear filtering
Optimization
Unscented Kalman filtering

ABSTRACT

The unscented Kalman filter (UKF) is a widely used nonlinear Gaussian filter. It has the potential to deal with highly nonlinear dynamic systems, while displaying computational cost of the same order of magnitude as that of the extended Kalman filter (EKF). The quality of the estimates produced by the UKF is dependent on the tuning of both the parameters that govern the unscented transform (UT) and the two noise covariance matrices of the system model. In this paper, the tuning of the UKF is framed as an optimization problem. The tuning problem is solved by a new stochastic search algorithm and by a standard model-based optimizer. The filters tuned with the proposed algorithm and with the standard model-based optimizer are numerically tested against other nonlinear Gaussian filters, including two UKF tuned with state-of-the-art tuning strategies. One of these strategies relies on online tuning and the other on offline tuning.

© 2017 Elsevier Ltd. All rights reserved.

1. Introduction

The UKF (Julier & Uhlmann, 1997) and the extended Kalman filter (EKF) (Maybeck, 1979) are two widely used approximate solutions to the problem of filtering nonlinear dynamical systems. The UKF has the potential to achieve better estimation performance than the EKF, while displaying the same order of computational complexity (Wan & van der Merwe, 2000). The heart of the UKF is the (scaled) unscented transform (UT) (Julier & Industries, 2002), which is aimed at propagating the mean and covariance of a random variable that undergoes a nonlinear transformation. For the UKF to work properly, it is thus necessary to tune the three scalar parameters of the UT. The state estimation performance of the UKF is also influenced by the tuning of the two noise covariance matrices of the system model used by the filter, since these noises can be used to model uncertainties about the true dynamic system (see Chapter 1 of Sarkka, 2013). The fact that the tuning of this important filter is such a difficult problem has motivated researchers to devise methods aimed at either tuning solely the UT, solely the noise covariance matrices, or both the UT and the noise covariance matrices. Let us now briefly review some of these methods.

In Julier, Uhlmann, and Durrant-Whyte (2000), for the formulation of the UT that features only the scaling parameter κ,[1] the user should set $\kappa = 3-n_x$, where n_x is the dimension of the state vector. In this paper, the UKF tuned with such rule is called UkfD.

In Sakai and Kuroda (2010), all three UT parameters plus the noise covariance matrices are obtained by gradient-free optimization. The optimization is carried out offline, implying no extra computational cost for the filter at runtime. The optimization process, though, demands the availability of highly accurate measurements of the state vector.

In Dunik, Simandl, and Straka (2010), Dunik, Simandl, and Straka (2012) and Straka, Dunik, and Simandl (2012), for the formulation of the UT that features only parameter κ, a value for this parameter is selected in real time from a user-defined discrete set of possible values. The UKF that results from this approach works according to the following basic steps. Each time a measurement is received by the filter, the measurement predictive moments are calculated for each one of the possible values of κ. The value of κ that yields the measurement predictive moments that maximize a given performance criterion, such as the likelihood of the measurement received by the filter, is chosen. The filter then uses the

☆ José Jaime da Cruz was partially supported by a CNPq grant. The material in this paper was not presented at any conference. This paper was recommended for publication in revised form by Associate Editor Antonio Vicino under the direction of Editor Torsten Söderström.

E-mail addresses: lascardua@ifes.edu.br, lascardua@gmail.com (L.A. Scardua), jaime@lac.usp.br (J.J. da Cruz).

http://dx.doi.org/10.1016/j.automatica.2017.01.008
0005-1098/© 2017 Elsevier Ltd. All rights reserved.

[1] This simplified UT is obtained when setting the other parameters of the scaled UT (Wan & van der Merwe, 2000) as $\alpha = 1$ and $\beta = 0$.

selected value of κ to perform its regular update and predict steps. The proposed approach thus imposes extra computational cost to the UKF at runtime. The amount of extra computation depends on the number of possible values of κ. This approach was further explored and refined in Straka, Dunik, and Simandl (2014), but the numerical results presented showed that the UKF enhanced with the proposed approach was still significantly slower than the regular UKF. In this paper, the UKF tuned with the approach described in Dunik et al. (2010), Dunik et al. (2012) and Straka et al. (2012) is called UkfK.

In Turner and Rasmussen (2010) and Turner and Rasmussen (2012), model-based optimization is used to pick values for the three UT parameters. The optimization process is guided by the maximization of the upper confidence bound (UCB) (Cox & John, 1997), which is a confidence-bound acquisition criterion (see Section 3). The only data needed from the original dynamic system are the measurements. The optimization is executed offline, so the approach does not increase the runtime computational cost of the UKF. The optimization process is performed by a gradient-based optimizer that relies on the derivative of the UCB. However, reliance on gradient information of the acquisition criteria may prevent the adoption of other relevant acquisition criteria. This is important because the choice of the acquisition criteria influences the optimization results. In this paper, the UKF tuned with this approach is called UkfO.

We approach the tuning of the UKF parameters as an optimization problem. The parameter space is composed by the UT scalar parameters and the main diagonal elements of the two noise covariance matrices of the system model used by the filter. A point in this space is represented by the vector $\boldsymbol{\theta} \in \Re^{n_\theta}$. The first dimension of $\boldsymbol{\theta}$ corresponds to the UT parameter α, the second to parameter β and the third to parameter κ. The other dimensions correspond to the diagonal elements of the noise covariance matrices. Two algorithms are used to perform the optimization.

The first algorithm is a standard model-based optimizer (Forrester, Sobester, & Keane, 2008) that uses a genetic algorithm (Goldberg, 1989) to maximize the UCB criterion. The UKF that results from this approach is here called UkfM. The second is new a tuning method based on stochastic search. This algorithm was used in Scardua and Cruz (2015) to tune solely the UT parameters. The UKF tuned with the second algorithm is called UkfP. For both algorithms, the tuning process is guided solely by the measurements obtained from the nonlinear dynamical system that is to be filtered.

To assess the impact of the tuning on the state estimation performance of the UKF, the two proposed tuning strategies (UkfM and UkfP) are numerically tested against the EKF, the cubature Kalman filter (CKF) (Arasaratnam & Haykin, 2009), the UkfD, the UkfK, and the UkfO. The EKF was chosen because it is a benchmark against which nonlinear filters are usually tested (Psiaki, 2013). UkfD and CKF both result from different tunings of the three UT scalar parameters, while UkfO and UkfK result from state-of-the-art tuning strategies. The numerical tests are performed on three well-known nonlinear filtering problems. To improve readability, we list some of the abbreviations used in the rest of the paper:

- UkfD—UKF tuned according to Julier et al. (2000);
- UkfP—UKF tuned with the proposed optimizer, described in Algorithm 3;
- UkfK—UKF tuned with the online approach of Dunik et al. (2010), Dunik et al. (2012) and Straka et al. (2012);
- UkfO—UKF tuned with the model-based approach of Turner and Rasmussen (2010) and Turner and Rasmussen (2012);
- UkfM—UKF tuned with the classical model-based optimizer, described in Algorithm 2.

The rest of this paper is organized as follows. Section 2 frames the tuning of the UKF parameters as an optimization problem. Section 3 provides background on model-based optimization and describes the UkfM algorithm. Section 4 describes algorithm UkfP. Section 5 analyzes the computational burden of Algorithm 3. Section 6 describes and analyzes the numerical experiments, and Section 7 presents the final comments.

2. Approaching the tuning of the UKF as an optimization problem

Consider the discrete-time nonlinear dynamic system

$$\begin{aligned} \mathbf{x}_k &= f(\mathbf{x}_{k-1}) + \mathbf{w}_{k-1}, \\ \mathbf{y}_k &= h(\mathbf{x}_k) + \mathbf{v}_k, \end{aligned} \tag{1}$$

where $\mathbf{x}_k \in \Re^{n_x}$ and $\mathbf{y}_k \in \Re^{n_y}$ are respectively the state of the dynamic system and its corresponding noisy measurement, $f(\cdot)$ and $h(\cdot)$ are known functions, $\mathbf{w}_{k-1}$ is the process noise at discrete time step $k-1$, and $\mathbf{v}_k$ is the measurement noise at discrete time step k. The distributions of the process and measurement noises are supposed to be Gaussian with zero means and unknown covariance matrices $\mathbf{Q}_{k-1}$ and $\mathbf{R}_k$. These assumptions are respectively written as $\mathbf{w}_{k-1} \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_{k-1})$ and $\mathbf{v}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_k)$. The initial state $\mathbf{x}_0$ and the two noises are considered to be independent random variables. Both noises are also assumed to be white.

We are interested in improving the quality of the UKF state estimates for (1), without increasing the computational burden of the filter during runtime.

The UKF is a Gaussian filter (Ito & Xiong, 2000). The general form of the Gaussian filter uses a Kalman-like structure to address the estimation task for system (1) (Ito & Xiong, 2000). Assuming that $\mathbf{m}_{k-1}$ and $\mathbf{P}_{k-1}$ are respectively the mean and covariance of the state estimation at time-step $k-1$, the equations of the general form are (Algorithm 6.3 of Sarkka, 2013):

2.1. Prediction

$$\begin{aligned} \mathbf{m}_k^- &= \int f(\mathbf{x}_{k-1}) \mathcal{N}(\mathbf{x}_{k-1} | \mathbf{m}_{k-1}, \mathbf{P}_{k-1}) d\mathbf{x}_{k-1}, \\ \mathbf{P}_k^- &= \int (f(\mathbf{x}_{k-1}) - \mathbf{m}_k^-)(f(\mathbf{x}_{k-1}) - \mathbf{m}_k^-)^T \\ &\quad \times \mathcal{N}(\mathbf{x}_{k-1} | \mathbf{m}_{k-1}, \mathbf{P}_{k-1}) d\mathbf{x}_{k-1} + \mathbf{Q}_{k-1}. \end{aligned} \tag{2}$$

2.2. Update

$$\begin{aligned} \boldsymbol{\mu}_k &= \int h(\mathbf{x}_k) \mathcal{N}(\mathbf{x}_k | \mathbf{m}_k^-, \mathbf{P}_k^-) d\mathbf{x}_k, \\ \mathbf{S}_k &= \int (h(\mathbf{x}_k) - \boldsymbol{\mu}_k)(h(\mathbf{x}_k) - \boldsymbol{\mu}_k)^T \mathcal{N}(\mathbf{x}_k | \mathbf{m}_k^-, \mathbf{P}_k^-) d\mathbf{x}_k + \mathbf{R}_k, \\ \mathbf{C}_k &= \int (\mathbf{x}_k - \mathbf{m}_k^-)(h(\mathbf{x}_k) - \boldsymbol{\mu}_k)^T \mathcal{N}(\mathbf{x}_k | \mathbf{m}_k^-, \mathbf{P}_k^-) d\mathbf{x}_k, \\ \mathbf{K}_k &= \mathbf{C}_k \mathbf{S}_k^{-1}, \\ \mathbf{m}_k &= \mathbf{m}_k^- + \mathbf{K}_k(\mathbf{y}_k - \boldsymbol{\mu}_k), \\ \mathbf{P}_k &= \mathbf{P}_k^- - \mathbf{K}_k \mathbf{S}_k \mathbf{K}_k^T. \end{aligned} \tag{3}$$

Different approaches to solving the moment integrals in (2) and (3) give rise to different Gaussian filters (Wu, Hu, Wu, & Hu, 2006). The UKF originates from using the UT to approximately calculate

those moment integrals. Let us now see how the UT parameters influence the quality of the estimates yielded by the UKF.

The UT approximately computes the mean and covariance of a random variable $\boldsymbol{y}$ that results from a nonlinear transformation $h(\cdot)$ of a Gaussian random variable $\boldsymbol{x} \sim \mathcal{N}(\bar{\boldsymbol{x}}, \boldsymbol{P}_x)$, with $\boldsymbol{x} \in \Re^{n_x}$. To this end, the dimensionless UT parameters $\{\alpha, \beta, \kappa\} \in \Re^+$ are used to calculate a set of $2n_x + 1$ deterministic points (sigma points) and weights. The sigma points are given by Wan and van der Merwe (2000)

$$\begin{aligned} \mathcal{X}_0 &= \bar{\boldsymbol{x}}, \\ \mathcal{X}_i &= \bar{\boldsymbol{x}} + \left(\sqrt{(n_x+\lambda)\mathbf{P}_x}\right)_i \quad i = 1, \ldots, n_x, \\ \mathcal{X}_i &= \bar{\boldsymbol{x}} - \left(\sqrt{(n_x+\lambda)\mathbf{P}_x}\right)_{(i-n_x)} \quad i = n_x+1, \ldots, 2n_x, \end{aligned} \tag{4}$$

where $(\cdot)_i$ denotes the ith column of the matrix, and λ is a scaling parameter given by

$$\lambda = \alpha^2(n_x + \kappa) - n_x. \tag{5}$$

The corresponding weights are

$$\begin{aligned} w_0^{(m)} &= \frac{\lambda}{n_x+\lambda}, \\ w_0^{(c)} &= \frac{\lambda}{n_x+\lambda} + (1-\alpha^2+\beta), \\ w_i^{(m)} &= w_i^{(c)} = \frac{1}{2(n_x+\lambda)} \quad i = 1, \ldots, 2n_x. \end{aligned} \tag{6}$$

Sigma points $\mathcal{X}_i$ are then used in the nonlinear transformation $h(\cdot)$, yielding

$$\mathcal{Y}_i = h(\mathcal{X}_i) \quad i = 0, \ldots, 2n_x. \tag{7}$$

Finally, the first two moments of $\boldsymbol{y}$ are approximated as

$$\begin{aligned} \bar{\boldsymbol{y}} &\approx \sum_{i=0}^{2n_x} w_i^{(m)} \mathcal{Y}_i, \\ \boldsymbol{P}_y &\approx \sum_{i=0}^{2n_x} w_i^{(c)} (\mathcal{Y}_i - \bar{\boldsymbol{y}})(\mathcal{Y}_i - \bar{\boldsymbol{y}})^T. \end{aligned} \tag{8}$$

The results obtained by the UT thus depend heavily on the values of α, β and κ. Accordingly, the quality of the UKF state estimates depends heavily on such values. Moreover, (2) and (3) show that noise matrices $\mathbf{Q}_{k-1}$ and $\mathbf{R}_k$ also play an important role in the quality of the state estimates yielded by the UKF. Proper tuning of both the UT parameters and the noise covariance matrices is thus essential to obtaining good UKF state estimation performance.

To approach the UKF tuning as an optimization problem, it was necessary to define a scalar goal function $g(\cdot)$ that could properly weight points of the UKF parameter domain. The weight of a given point should be an indication of the state estimation performance of the UKF when tuned with that point. The tuning process would then consist in the maximization of a scalar goal function $g(\boldsymbol{\theta})$, where $\boldsymbol{\theta}$ is a point in the parameter space.

The tuning process described in this article adopts the goal function used in Turner and Rasmussen (2010). This goal function maximizes the likelihood of the measurements that serve as training data. For a given point $\boldsymbol{\theta}$, this goal function runs a UKF tuned with $\boldsymbol{\theta}$ on the set $\mathbf{Y} = \{\mathbf{Y}_i\}$ of n_Y independent time series $\mathbf{Y}_i = \{\mathbf{y}_i(1), \mathbf{y}_i(2), \ldots, \mathbf{y}_i(T)\}$, where $\mathbf{y}_i(k)$, $k \in \{1, \ldots, T\}$, is a noisy measurement of the dynamic system state taken at discrete time step k.

For each $\mathbf{Y}_i$, the means and covariances of the measurement predictions yielded by the UKF tuned with $\boldsymbol{\theta}$ are used to calculate the log of the conditional likelihood $p(\mathbf{Y}_i|\boldsymbol{\theta})$, which is given by

$$\log\, p(\mathbf{Y}_i|\boldsymbol{\theta}) = \frac{\sum_{k=1}^{T} \log\left(\mathcal{N}(\mathbf{y}_i(k)|\hat{\mathbf{y}}_i(k), \mathbf{S}_i(k))\right)}{T}, \tag{9}$$

where $\hat{\mathbf{y}}_i(k)$ and $\mathbf{S}_i(k)$ are respectively the mean and covariance of the measurement estimates yielded by the UKF tuned with $\boldsymbol{\theta}$, at discrete time instant k. The sequence of calculations performed by the goal function is shown in Algorithm 1. The goal function receives the UT parameters $\boldsymbol{\theta}$, the training time series $\mathbf{Y} = \{\mathbf{Y}_1, \ldots, \mathbf{Y}_{n_Y}\}$, and both the mean $\mathbf{M}_0$ and covariance $\mathbf{P}_0$ of the initial state estimate. It then returns the conditional log-likelihood of $\mathbf{Y}$.

Algorithm 1 Goal Function

```
procedure G(θ, Y, M_0, P_0)
    for each Y_i ∈ Y do
        [Ŷ_i, S_i] ← UKF(θ, Y_i, M_0, P_0)  ▷ Run the UKF tuned with θ
        loglik(Y_i) = log p(Y_i, θ)  ▷ The conditional log-likelihood of Y_i is given by (9)
    w_θ ← Σ_1^{n_Y} loglik(Y_i)  ▷ Calculate the conditional log-likelihood of Y
    return w_θ
```

The maximization of the log-likelihood of the training data $\mathbf{Y}$ does not guarantee the maximization of the state estimation performance of the filter, since no information about the true states that gave rise to the measurements is used (Scardua & Cruz, 2015). It would be more convenient to maximize the likelihood of the true states that gave rise to the measurements, but in many cases it would be costly or even impossible to obtain such ground-truth data. Fortunately, the results described in Section 6 show that it is possible to significantly improve the performance of the UKF without having to obtain ground-truth data.

It is also important to highlight that the performance of the tuned UKF on previously unseen data is dependent on how well the training data represents the situations the filter would encounter in the unseen data. The more representative the training data is, the better the tuned filter will perform in unseen data.

3. Model based optimization

Two of the tuning algorithms tested in this paper, UkfO and UkfM, make use of the model-based optimization (Forrester et al., 2008) strategy. Model-based optimization is usually adopted in situations where the goal function $g(\cdot)$ is a costly-to-evaluate black box function. The general strategy of the model-based optimization approach is to build a cheap computational model of the goal function, and use this model as a surrogate to seek the actual goal function maxima. To this end, model-based optimization relies on probabilistic meta-models, such as Gaussian Processes (GP) (Rasmussen & Williams, 2006), to build the surrogate model.

GP models are obtained by fitting a GP to the set $\mathcal{D} = \{(\boldsymbol{\theta}_1, g(\boldsymbol{\theta}_1)), \ldots, (\boldsymbol{\theta}_{n_D}, g(\boldsymbol{\theta}_{n_D}))\}$, where $g(\boldsymbol{\theta}_i)$ is the value of the goal function at point $\boldsymbol{\theta}_i$. Once trained, the model can be used to predict the value of the goal function at input points that do not belong to $\mathcal{D}$. To make predictions, GP models rely on the idea that two points in the input space that are close to each other are likely to have similar values of the goal function. Hence, the goal function values at

training points that are near a previously unseen point are considered to be informative about the value of the goal function at that point.

The GP framework uses covariance functions to relate the values of the goal function at two input points to the distance between these points. One simple covariance function is the *squared exponential* (SE), which is given by

$$\mathrm{cov}(g(\boldsymbol{\theta}_i), g(\boldsymbol{\theta}_j)) = \exp\left(-\frac{1}{2}|\boldsymbol{\theta}_i - \boldsymbol{\theta}_j|^2\right). \tag{10}$$

For the SE function, the covariance between the goal function values at two input points approaches 1 as the corresponding input points get close to each other, and decreases towards 0 as the distance between them increases. GP models make use of this property to associate covariances to their predictions. The closer the point of the prediction is to the training points, the more confident is the prediction (the lower is the covariance associated with the prediction). In model-based optimization, the initial set of samples $\mathcal{D}$ is not usually sufficient to produce a good model of the goal function. Improving the accuracy of the GP model demands acquiring more training data.

Sequential model-based optimization (SMBO) algorithms iterate between fitting the model and gathering additional training points based on the predictions yielded by the model. As evaluations of the goal function are assumed to be costly, it is necessary to choose carefully where to next evaluate the goal function. This is achieved by using the predictions of the surrogate to find the point $\boldsymbol{\theta}_l$ of the input space that maximizes a given (acquisition) criterion. The goal function is evaluated at $\boldsymbol{\theta}_l$, and the resulting data pair $(\boldsymbol{\theta}_l, g(\boldsymbol{\theta}_l))$ is added to $\mathcal{D}$. The model is fit to the enlarged $\mathcal{D}$ and the optimizer uses the upgraded model to search for a new point where to evaluate the goal function. This iterative process is usually limited by the optimization budget, which is the maximum allowed number of evaluations of the goal function. In summary, the general steps of SMBO algorithms (Jones, Schonlau, & Welch, 1998) are shown in Algorithm 2.

Algorithm 2 SMBO General Steps

1: **procedure** SMBO
2: Build an initial stochastic model of the goal function (the surrogate model).
3: Use the surrogate model to seek for the location of the input space that maximizes the chosen acquisition criterion.
4: Add an evaluation of the goal function at the location found in Step 3.
5: Update the surrogate model using the new data point.
6: If the stopping criterion (such as the maximum number of evaluations of the goal function) has not been met, go to Step 3.

There are a number of acquisition criteria (Brochu, Cora, & de Freitas, 2009) to guide the search for the point of the input space where to next evaluate the goal function. These criteria usually make use of the fact that predictions yielded by GP models are composed of a mean value (the predicted value) and a variance. Acquisition criteria can use the mean and the variance of the predictions to balance the conflicting strategies of searching near sample points that display higher values of the goal function (exploitation), and searching in regions of the input space where there are fewer samples (exploration). Exploitation is aimed at achieving faster convergence, while exploration is necessary to minimize the chance of getting stuck at local maxima. One acquisition criterion that explicitly seeks to balance exploration and exploitation is the upper confidence bound (UCB) (Cox & John, 1992, 1997). The UCB criterion at a point $\boldsymbol{\theta}$ is given by

$$\mathrm{UCB}(\boldsymbol{\theta}) = \mu(\boldsymbol{\theta}) + A\sigma(\boldsymbol{\theta}), \tag{11}$$

where $A > 0$ is a scalar value, while $\mu(\boldsymbol{\theta})$ and $\sigma(\boldsymbol{\theta})$ are respectively the mean and standard deviation of the prediction yielded by the model at $\boldsymbol{\theta}$. The value of A can significantly influence the optimization results (Jones, 2001). A method for automatically choosing A is described in Srinivas, Krause, Kakade, and Seeger (2010), but it is common practice to set $A = 2$.

The general steps of the tuning algorithm that yields UkfM are given by Algorithm 2, with step 3 implemented by a genetic algorithm. In this article, the genetic algorithm is implemented with the *ga.m* function from Matlab R2016a, with default options.

4. Stochastic tuning of the UKF

The goal function given in Algorithm 1 can be seen as a black-box function, suggesting the adoption of gradient-free optimization algorithms, such as pattern search (Hooke & Jeeves, 1961) or genetic algorithms. Nonetheless, existing gradient-free optimizers may require a large number of function evaluations, and Algorithm 1 may be costly to evaluate in a number of situations, such as when the time series that comprise the training data are long or when the dynamic model (1) takes significant computational time to run.

The idea is thus to conceive an easy-to-implement direct search algorithm able to find a good tuning of the UKF parameters without the need to perform too many evaluations of the goal function. It is considered to be good a tuning that outperforms the EKF, the UkfD and the CKF in terms of state estimation performance, as measured by the criteria described in Section 6.

The goal of finding a good tuning can be achieved by assuming that the goal function shows some degree of smoothness. It is then reasonable to expect that two points in the input space (of the goal function) that are close to each other are likely to have similar values of the goal function. Hence, the value of the goal function at a given point can be estimated by the values of the goal function at points that are near. This property means that close to points where the value of the goal function is high there are points where the value of the goal function is similarly high, or maybe even higher.

As a result, an optimization algorithm that searches close to points that display high values of the goal function described by Algorithm 1 would have the potential to find points where this goal function displays even higher values. Nonetheless, to minimize the chance of getting stuck at poor local maxima, the optimizer must implement some sort of exploration of the input space.

To balance the conflicting goals of searching near points where the goal function is known to be high (exploitation of what is already known) and searching in regions of the input space where there are few or no samples at all (exploration of the input space), an optimization algorithm could evaluate the goal function at points sampled from a given probability distribution. This probability distribution should be conceived in a way that it would produce points located near and also points located not so near existing samples. The former points would allow the algorithm to exploit what is already known about the shape of the goal function, while the latter points would allow the algorithm to explore the input space.

We thus propose tackling the tuning problem with a stochastic search algorithm that works as follows. It starts with a set $\boldsymbol{\Theta}$ of points that act as candidate solutions to the tuning problem. At each iteration, the goal function is evaluated at the candidate points, and only the points that yield the N_s highest values of the goal function are kept in $\boldsymbol{\Theta}$. The values of the goal function corresponding to the points in the new $\boldsymbol{\Theta}$ are then normalized to interval [0, 1], in a way that the points in $\boldsymbol{\Theta}$ and the normalized values form a discrete probability distribution (the normalized values are henceforth called weights). A population $\boldsymbol{\Theta}_{best}$ is then formed by randomly sampling N_s points from this discrete distribution (in the numerical experiments of Section 6, the resampling

is performed according to the stratified resampling strategy Douc & Cappe, 2005).

The resampling steadily increases the number of copies of points with high weights and decreases the number of copies of points with low weights. This can be seen as a way to exploit the information conveyed by the existing points, but it does not bring any new information to the optimization process. To bring new information to the optimization process, the algorithm samples N_s new points from the distribution

$$p(\boldsymbol{\theta}_{new}) = \mathcal{N}(\mathbf{M}_{best}, \mathbf{S}_{best}), \tag{12}$$

where $\mathbf{M}_{best}$ and $\mathbf{S}_{best}$ are respectively the mean and covariance of $\boldsymbol{\Theta}_{best}$. The normal distribution was chosen because it is completely specified by $\mathbf{M}_{best}$ and $\mathbf{S}_{best}$, making it easy to obtain new samples.

Probability distribution (12) is also interesting because it can yield samples located within varying distances of the best existing points. Samples that are close to existing points allow the tuning algorithm to exploit existing information about the goal function, while points that are farther away allow the algorithm to explore the input space.

The goal function is evaluated at $\boldsymbol{\theta}_{new}$, providing them with weights. The new points and their weights are then added to the current population of points, thus bringing new information to the optimization process.

A more systematic description of these general tuning steps is given in Algorithm 3, which features a number of parameters. $\mathbf{Y}^{train} = \{\mathbf{Y}_1^{train}, \ldots, \mathbf{Y}_{N_{tr}}^{train}\}$ is the set of time series demanded by the goal function given by Algorithm 1. N_{eval} is the budget of evaluations of the goal function. $\boldsymbol{\Delta}_\theta$ is the range of possible values for the points. $\boldsymbol{\Theta} = \{\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_N\}$ is the set of initial points. N_s is the number of new points generated at each iteration. $\mathbf{M}_0, \mathbf{P}_0$ are respectively the mean and covariance of the initial state (see Algorithm 1).

Algorithm 3 Stochastic UKF Tuning

1: **procedure** UKFP($\mathbf{Y}^{train}$, N_{eval}, Δ_θ, Θ, N_s, $\mathbf{M}_0$, $\mathbf{P}_0$)
2: $\quad\mathbf{W}_\Theta \leftarrow g(\Theta, \mathbf{Y}^{train}, \mathbf{M}_0, \mathbf{P}_0)$ ▷ Evaluate the goal function at each point in Θ
3: $\quad\Theta_{best} \leftarrow \Theta$
4: $\quad$**for** $i = 1$ to $\frac{\mathrm{N}_{eval}}{N_s}$ **do**
5: $\quad\quad\mathbf{M}_{best} \leftarrow$ Mean of points in Θ_{best}.
6: $\quad\quad\mathbf{P}_{best} \leftarrow$ Covariance of points in Θ_{best}.
7: $\quad\quad$**if** $\mathbf{P}_{best}$ not positive definite **then**
8: $\quad\quad\quad\mathbf{M}_{best} \leftarrow$ best point in Θ
9: $\quad\quad\quad\mathbf{P}_{best} \leftarrow$ identity matrix
10: $\quad\quad\Theta_{new} \leftarrow$ Sample N_s points from $\mathcal{N}(\mathbf{M}_{best}, \mathbf{P}_{best})$.
11: $\quad\quad\Theta_{new} \leftarrow \mathrm{trim}(\Theta_{new}, \Delta_\theta)$ ▷ Trim the coordinate values of the points Θ_{new} to enforce the limits defined by Δ_θ.
12: $\quad\quad\mathbf{W}_{new} \leftarrow g(\Theta_{new}, \mathbf{Y}^{train}, \mathbf{M}_0, \mathbf{P}_0)$ ▷ Weights for points in Θ_{new}.
13: $\quad\quad\Theta \leftarrow \{\Theta; \Theta_{new}\}$ ▷ Append the new points to Θ.
14: $\quad\quad\mathbf{W}_\Theta \leftarrow \{\mathbf{W}_\Theta; \mathbf{W}_{new}\}$ ▷ Append the new weights to $\mathbf{W}_\Theta$.
15: $\quad\quad\mathbf{W}_\theta \leftarrow$ Keep only the N_s highest weights in $\mathbf{W}_\theta$.
16: $\quad\quad\Theta \leftarrow$ Keep only the points corresponding to $\mathbf{W}_\theta$.
17: $\quad\quad\Theta_{best} \leftarrow$ resample N_s points from a discrete distribution formed with Θ and $\mathbf{W}_\theta$
18: $\quad\boldsymbol{\theta}^\star \leftarrow$ Return the point associated with the highest weight.

5. Computational burden

Algorithm 3 performs offline tuning of the UKF. As a result, it does not impose any extra computational cost to the filter at runtime. The model-based tuning algorithms that yield UkfO and UkfM also do not incur in any extra computational cost to the UKF at runtime, but are much more computationally complex than Algorithm 3. For instance, at each iteration, the model-based tuning algorithms fit GP models to an increasing number of training pairs $(\boldsymbol{\theta}, g(\boldsymbol{\theta}))$. Computational complexities of a GP are $\mathcal{O}(n^3)$ for learning and $\mathcal{O}(n^2)$ for prediction, where n is the number of training pairs (Cao, Brubaker, Fleet, & Hertzmann, 2015). For large n, these computational complexities would represent considerable computational costs. Calculations performed by Algorithm 3 are simpler. Moreover, steps 15 and 16 imply that each iteration of Algorithm 3 always works with N_s candidate points, thus keeping the computational cost of each iteration constant throughout the optimization process.

6. Numerical experiments

Dynamic systems Kitagawa, Sinusoid and Bearings Only are used to numerically compare the state estimation quality of the filters.

6.1. Kitagawa

Variations of the Kitagawa (Kitagawa, 1996) model have been used as benchmark for nonlinear filters (Gordon, Salmond, & Smith, 1993; Turner & Rasmussen, 2010, 2012). The model equations are

$$\begin{aligned} x_{k+1} &= 0.5x_k + \frac{25x_k}{(1+x_k^2)} + w_k, \quad w_k \sim \mathcal{N}(0, 0.2^2) \\ y_k &= 5\sin(2x_k) + v_k, \quad v_k \sim \mathcal{N}(0, 0.01^2). \end{aligned} \tag{13}$$

The initial state estimate for generating both the training and test data was $\mathbf{x}_0 \sim \mathcal{N}(0, 0.5^2)$.

6.2. Sinusoid

The Sinusoid problem (Turner & Rasmussen, 2010, 2012) is given by

$$\begin{aligned} x_{k+1} &= 3\ \sin(x_k) + w_k, \quad w_k \sim \mathcal{N}(0, 0.1^2) \\ y_k &= \sigma(x_k/3) + v_k, \quad v_k \sim \mathcal{N}(0, 0.1^2), \end{aligned} \tag{14}$$

where $\sigma(\cdot)$ represents the logistic sigmoid. The initial state distribution for both the training and test data was $\mathbf{x}_0 \sim \mathcal{N}(0, 1^2)$.

6.3. Bearings Only

Different versions of the Bearings Only are also popular (Bar-Shalom, Kirubarajan, & Li, 2002; Dunik et al., 2012). The model represents the problem of tracking a moving target based solely on measurements provided by one angular sensor. In this paper, the system state is the position of the target in the Cartesian plane. The model equations are

$$\begin{aligned} \mathbf{x}_{k+1} &= \begin{bmatrix} 0.9 & 0 \\ 0 & 1 \end{bmatrix} \mathbf{x}_k + \mathbf{w}_k, \quad \mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}) \\ y_k &= \tan^{-1}\left(\frac{x_{2,k} - \sin(k)}{x_{1,k} - \cos(k)}\right) + v_k, \quad v_k \sim \mathcal{N}(0, R), \end{aligned} \tag{15}$$

where

$$\mathbf{Q} = \begin{bmatrix} 0.1 & 0.01 \\ 0.01 & 0.1 \end{bmatrix}, \qquad R = 0.025. \tag{16}$$

The distribution of the initial state for both the training and test data was

$$\mathbf{x}_0 \sim \mathcal{N}\left(\begin{bmatrix} 20 \\ 5 \end{bmatrix}, \begin{bmatrix} 0.1 & 0 \\ 0 & 0.1 \end{bmatrix}\right). \tag{17}$$

6.4. Tuning the UKF

The tunings for UkfO, UkfM and UkfP were obtained by feeding the optimizers respectively described in Turner (2010), in Algorithm 2 and in Algorithm 3 with the necessary parameters. These parameters are

- $\mathbf{Y}^{train} = \{\mathbf{Y}_1^{train}, \ldots, \mathbf{Y}_{N_{tr}}^{train}\}$, which was comprised of $N_{tr} = 5$ time series generated by Monte Carlo simulation of the nonlinear system. For Kitagawa and Sinusoid, the time series were 100 time-steps long, while for Bearings Only, they were 1000 time-steps long;
- N_{eval}, which was set as 100 evaluations of the goal function;
- $\mathbf{\Delta}_\theta$, defined as

  $$0.01 \le \alpha \le 4, \qquad 0 \le \beta \le 4, \qquad 0 \le \kappa \le 5$$
  $$0.1\mathrm{Q}_{ii} \le \mathrm{Qf}_{ii} \le 10\mathrm{Q}_{ii}, \quad i = \{1, \ldots, n_x\}$$
  $$0.1\mathrm{R}_{jj} \le \mathrm{Rf}_{jj} \le 10\mathrm{R}_{jj}, \quad j = \{1, \ldots, n_y\}$$

  where Q_{ii} and R_{jj} stand for the diagonal elements of the noise covariance matrices used to generate the test data, while Qf_{ii} and Rf_{jj} are the diagonal elements of the noise covariance matrices used in the filters;
- $\mathbf{\Theta}$, generated as a grid in the space defined by $\mathbf{\Delta}_\theta$. For problems Kitagawa and Sinusoid, the initial set of samples consisted of 32 points. For the Bearings Only problem, the initial set of samples was comprised of 64 points. The initial points were calculated by the same function that calculates the initial points in the experiments described in Turner and Rasmussen (2010); and
- $\mathbf{M}_0$ and $\mathbf{P}_0$, which define the probability distribution of the initial state.

Algorithm 3 also received parameter $N_s = 10$, which is not demanded by the other tuning algorithms. Filter UkfK receives as parameter the range of possible values for κ. In the numerical experiments, this range was the discrete set $\{0, 0.1, 0.2, \ldots, 4.8, 4.9, 5\}$.

6.5. Performance criteria

The performance criteria are the root mean squared error (RMSE) and the mean absolute error (MAE) of the estimated states (Chai & Draxler, 2014). The MAE measures the mean magnitude of the absolute errors, meaning that all errors equally influence the criterion value. RMSE is a quadratic criterion, entailing that bigger errors have more influence on the final value of the criterion than smaller errors. The two criteria thus provide different views of the estimation performances. For both, smaller values mean better performance.

Box plots of the squared state estimation errors are also shown. They help visualize the differences in the performances of the filters. In general, worse performances are linked to more outliers, and/or higher medians and/or wider interquartile ranges.

6.6. Test data

The data used to numerically test the filters, $\mathbf{Y}^{test} = \{\mathbf{Y}_1^{test}, \ldots, \mathbf{Y}_{N_{ts}}^{test}\}$, was comprised of $N_{ts} = 500$ independent time series generated by Monte Carlo simulation of the nonlinear system. Each time series was 100 time steps long. The testing of the filters consisted in assessing the quality of the state estimates yielded by the filters on the true states corresponding to $\mathbf{Y}^{test}$.

6.7. Results

The state estimation errors for all three benchmark problems are shown in Tables 1–3. In all cases, EKF, CKF, UkfD and UkfK were run with the same noise covariance matrices used to generate the test data, which favors the performances of these filters.

The parameters found by the offline optimization algorithms are shown in Tables 4–6.

The variability of the squared state estimation errors of the filters are shown in Figs. 1–3.

Table 1
Performance metrics of the filters in Kitagawa.

	RMSE	MAE	Mean time (s)
UkfD	1.167	0.248	0.009
CKF	3.901	1.107	0.009
UkfO	2.124	0.318	0.009
UkfK	1.142	0.270	0.077
EKF	5.813	1.957	0.001
UkfP	0.898	0.201	0.009
UkfM	0.884	0.193	0.009

Table 2
Performance metrics of the filters in Sinusoid.

	RMSE	MAE	Mean time (s)
UkfD	0.980	0.781	0.009
CKF	1.201	0.889	0.009
UkfO	0.854	0.675	0.009
UkfK	1.191	0.909	0.073
EKF	1.241	0.938	0.001
UkfP	0.873	0.689	0.009
UkfM	1.416	1.056	0.009

Table 3
Performance metrics of the filters in Bearings Only.

	RMSE	MAE	Mean time (s)
UkfD	3.263	2.371	0.016
CKF	4.153	3.135	0.016
UkfO	2.478	2.125	0.016
UkfK	1.428	1.325	0.120
EKF	7.230	7.490	0.003
UkfP	1.989	1.626	0.016
UkfM	2.819	2.579	0.016

Table 4
Parameters of the tuned filters for the Kitagawa problem.

	α	β	κ	Q_{11}	R_{11}
UkfM	0.3284	4.0000	2.5015	0.0211	0.0001
UkfP	0.4150	1.0469	2.6261	0.0515	0.0006
UkfO	1.0075	1.0000	1.2500	0.1030	0.0008

Table 5
Parameters of the tuned filters for the Sinusoid problem.

	α	β	κ	Q_{11}	R_{11}
UkfM	1.9313	0.0001	1.4564	0.0560	0.0229
UkfP	1.3937	0.0000	0.4058	0.0080	0.0132
UkfO	1.9964	0.3493	0.2891	0.0429	0.0116

Table 6
Parameters of the tuned filters for the Bearings Only problem.

	α	β	κ	Q_{11}	Q_{22}	R_{11}
UkfM	1.0075	1.0000	1.2500	0.2575	0.7525	0.1881
UkfP	0.8472	2.3538	1.8219	0.1024	0.2822	0.0985
UkfO	0.7119	0.8084	2.4314	0.1387	0.7802	0.1246

6.8. Analysis

In terms of state estimation quality, Tables 1–3 show that UkfP was able to outperform the non-tuned filters (CKF and EKF) in all three test problems. Despite the fact that both CKF and EKF use

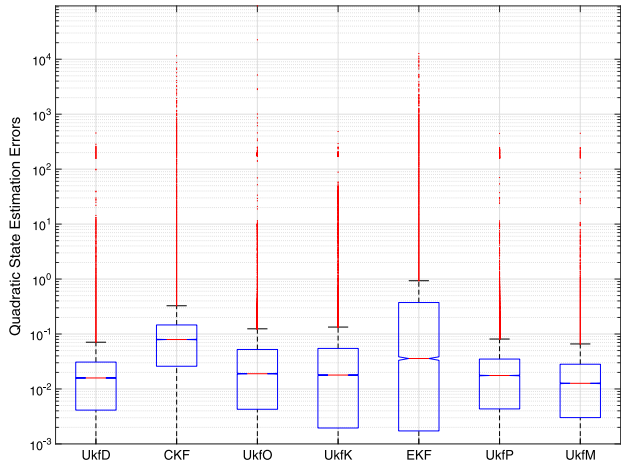

Fig. 1. Variability of the squared state estimation errors for Kitagawa.

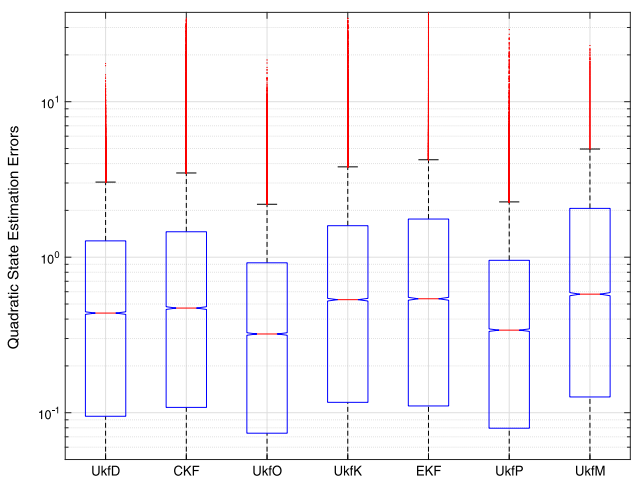

Fig. 2. Variability of the squared state estimation errors for Sinusoid.

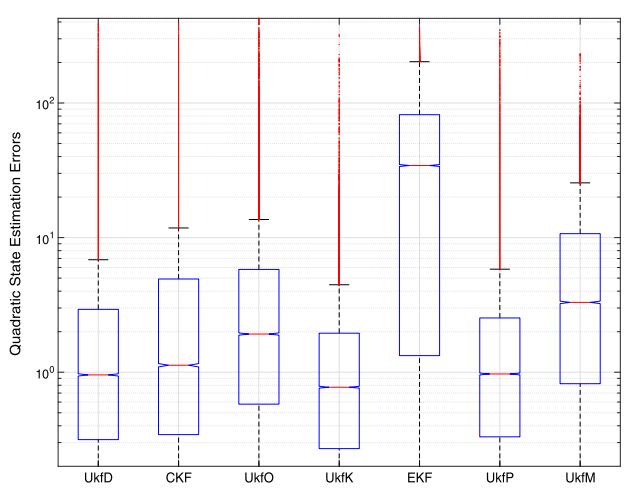

Fig. 3. Variability of the squared state estimation errors for Bearings Only.

the true noise covariance matrices, they were always among the worst performing filters. At this point it is important to address an apparent contradiction. The poor performance of CKF depicted in Table 3 may at first seem to be in contradiction to what is shown in Fig. 3, since the CKF box plot is positioned clearly below the UkfO box plot, and both box plots seem to display a similar number of outliers spread across similar ranges of values. This apparent contradiction is due to the fact that the log scale makes it difficult to see clearly that the CKF box plot displays almost twice as many outliers as the UkfO box plot.

Considering the tuned filters, UkfP was able to deliver the most consistent performance. It was second to UkfM in Kitagawa and second to UkfO in Sinusoid. In both cases, UkfP displayed almost the same state estimation performance as that of the best performing filter. In Bearings Only, UkfP was second to the online-tuned UkfK. The more consistent state estimation performance achieved by UkfP can also be seen in Figs. 1–3, since the variability of UkfP is always on par with that of the best performing filter.

The other tuned filters did not perform as consistently as UkfP. UkfO displayed poorer performances both in Kitagawa and Bearings Only, while UkfM displayed poorer performances both in Sinusoid and Bearings Only. UkfK should have displayed better performances in Kitagawa and Sinusoid, since it uses the true noise covariance matrices. UkfD, which also uses the true noise covariance matrices, performed poorly in Bearings Only.

In terms of computational performance, the mean execution times shown in Tables 1–3 provide an objective measure of the significant computational cost imposed by the online tuning strategy of UkfK, possibly limiting its applicability. As expected, the offline tuning processes that rendered filters UkfP, UkfM and UkfO do not impose extra computational cost to the filters at runtime.

7. Conclusion

In this paper, the state estimation performance of the UKF has been improved by proper tuning of both the UT parameters and the two noise covariance matrices of the dynamic system model. To this end, two optimization algorithms were used, a new stochastic tuning algorithm and a standard model-based optimizer. The optimization was performed offline, allowing the UKF to retain its low original computational complexity.

The new tuning algorithm, despite being simple to implement, managed to improve the state estimation performance of the UKF on three commonly used nonlinear benchmark filtering problems, having consistently outperformed the non-tuned UKF and outperforming or performing on par with the other tuned UKF. The numerical comparisons included state-of-the-art offline and online tuning algorithms, providing a picture of the current research in the area of UKF tuning.

Except for the EKF, the only difference between the compared filters was the tuning of the UKF parameters. The numerical results thus indicate how important it is to properly tune the UKF parameters for the UKF state estimation performance.

References

Arasaratnam, I., & Haykin, Simon (2009). Cubature Kalman filters. *IEEE Transactions on Automatic Control*, *54*(6), 1254–1269.

Bar-Shalom, Yaakov, Kirubarajan, Thiagalingam, & Li, X.-Rong (2002). *Estimation with applications to tracking and navigation*. New York, NY, USA: John Wiley & Sons, Inc.

Brochu, Eric, Cora, Vlad M., & de Freitas, Nando (2009). *A tutorial on Bayesian optimization of expensive cost functions, with application to active user modeling and hierarchical reinforcement learning. Technical Report TR-2009-23*. Department of Computer Science, University of British Columbia.

Cao, Y., Brubaker, M. A., Fleet, D. J., & Hertzmann, A. (2015). Efficient optimization for sparse gaussian process regression. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *37*(12), 2415–2427.

Chai, T., & Draxler, R. R. (2014). Root mean square error (rmse) or mean absolute error (mae)? arguments against avoiding rmse in the literature. *Geoscientific Model Development*, *7*(3), 1247–1250.

Cox, D.D., & John, S. (1992). A statistical method for global optimization. In *IEEE international conference on systems, man and cybernetics, 1992*, vol. 2, (pp. 1241–1246).

Cox, Dennis D., & John, Susan (1997). SDO: A statistical method for global optimization. In Natalia M. Aleksandrov, & M. Y. Hussaini (Eds.), *Multidisciplinary design optimization: state-of-the-art* (pp. 315–329). SIAM.

Douc, R., & Cappe, O. (2005). Comparison of resampling schemes for particle filtering. In *Proceedings of the 4th international symposium on image and signal processing and analysis, 2005*. (pp. 64–69).

Dunik, J., Simandl, M., & Straka, O. (2010). Adaptive choice of scaling parameter in derivative-free local filters. In *2010 13th conference on information fusion (FUSION)*, (pp. 1–8).

Dunik, Jindrich, Simandl, Miroslav, & Straka, Ondrej (2012). Unscented Kalman filter: Aspects and adaptive setting of scaling parameter. *IEEE Transactions on Automatic Control*, *57*(9), 2411–2416.
Forrester, Alexander I. J., Sobester, Andras, & Keane, Andy J. (2008). *Engineering design via surrogate modelling—A practical guide*. Wiley.
Goldberg, David E. (1989). *Genetic algorithms in search, optimization, and machine learning* (1st ed.). Boston, MA, USA: Addison-Wesley Professional.
Gordon, N. J., Salmond, D. J., & Smith, A. F. M. (1993). Novel approach to nonlinear/non-Gaussian Bayesian state estimation. *IEE Proceedings F Radar and Signal Processing*, *140*(2), 107–113.
Hooke, Robert, & Jeeves, T. A. (1961). direct search solution of numerical and statistical problems. *Journal of the ACM*, *8*(2), 212–229.
Ito, K., & Xiong, Kaiqi (2000). Gaussian filters for nonlinear filtering problems. *IEEE Transactions on Automatic Control*, *45*(5), 910–927.
Jones, DonaldR. (2001). A taxonomy of global optimization methods based on response surfaces. *Journal of Global Optimization*, *21*(4), 345–383.
Jones, Donald R., Schonlau, Matthias, & Welch, William J. (1998). Efficient global optimization of expensive black-box functions. *Journal of Global Optimization*, *13*(4), 455–492.
Julier, S.J., & Industries, Idak (2002). The scaled unscented transformation. In *in Proc. IEEE Amer. control conf*, (pp. 4555–4559).
Julier, S. J., Uhlmann, J. K., & Durrant-Whyte, H. F. (2000). A new method for the nonlinear transformation of means and covariances in filters and estimators. *IEEE Transactions on Automatic Control*, *45*(3), 477–482.
Julier, Simon J., & Uhlmann, Jeffrey K. (1997). A new extension of the Kalman filter to nonlinear systems. In *Proc. of aerosense: the 11th int. symp. on aerospace/defense sensing, simulations and controls*, (pp. 182–193).
Kitagawa, Genshiro (1996). Monte Carlo filter and smoother for non-Gaussian nonlinear state space models. *Journal of Computational and Graphical Statistics*, *5*(1), 1–25.
Maybeck, Peter S. (1979). *Mathematics in science and engineering*: *Vol. 1. Stochastic models, estimation and control*. Academic Press.
Psiaki, Mark. L. (2013). The blind tricyclist problem and a comparative study of nonlinear filters: A challenging benchmark for evaluating nonlinear estimation methods. *IEEE Control Systems*, *33*(3), 40–54.
Rasmussen, C. E., & Williams, C. K. I. (2006). *Adaptive computation and machine learning*: *Vol. 1. Gaussian processes for machine learning*. Cambridge, MA, USA: MIT Press.
Sakai, A., & Kuroda, Y. (2010). Discriminatively trained unscented Kalman filter for mobile robot localization. *Journal of Advanced Research in Mechanical Engineering (JARME)*, *1*(3), 153–161.
Sarkka, Simo (2013). *Bayesian filtering and smoothing*. Cambridge University Press.
Scardua, Leonardo Azevedo, & Cruz, José Jaime (2015). Particle-based tuning of the unscented kalman filter. *Journal of Control, Automation and Electrical Systems*, *27*(1), 10–18.
Srinivas, Niranjan, Krause, Andreas, Kakade, Sham, & Seeger, Matthias (2010). Gaussian process optimization in the bandit setting: No regret and experimental design. In Johannes Furnkranz, & Thorsten Joachims (Eds.), *Proceedings of the 27th international conference on machine learning*, (ICML-10). (pp. 1015–1022). Haifa, Israel: Omnipress.
Straka, O., Dunik, J., & Simandl, M. (2012). Scaling parameter in unscented transform: Analysis and specification. In *American control conference (ACC), 2012*, june, (pp. 5550–5555).
Straka, Ondrej, Dunik, Jindrich, & Simandl, Miroslav (2014). Unscented Kalman filter with advanced adaptation of scaling parameter. *Automatica*, *50*(10), 2657–2664.
Turner, Ryan Code for mlsp 2010. 2011. http://mlg.eng.cam.ac.uk/rdturner/code.html (Accessed 4 November 14).
Turner, Ryan, & Rasmussen, Carl Edward (2010). Model based learning of sigma points in unscented Kalman filtering. In *Machine learning for signal processing (MLSP'10)*, Kittila, Finland, August, (pp. 178–183).
Turner, Ryan, & Rasmussen, Carl Edward (2012). Model based learning of sigma points in unscented Kalman filtering. *Neurocomput*, *80*, 47–53.
Wan, E.A., & van der Merwe, R. (2000). The unscented Kalman filter for nonlinear estimation. In *Adaptive systems for signal processing, communications, and control symposium 2000. AS-SPCC. The IEEE 2000*, (pp. 153–158).
Wu, Yuanxin, Hu, Dewen, Wu, Meiping, & Hu, Xiaoping (2006). A numerical-integration perspective on Gaussian filters. *IEEE Transactions on Signal Processing*, *54*(8), 2910–2921.

Leonardo Azevedo Scardua received the B.S.E.E. degree in 1989 and the M.Sc. degree in 1996, both from the Federal University of Espírito Santo, Brazil, and the D.Sc. degree from the University of São Paulo Brazil, in 2015. He has engineering experience with software systems for the railway industry, having developed train dispatching systems for more than a decade. He is now an associate professor in the Control Engineering Department at the Federal Institute of Technology of Espírito Santo, Brazil. His current research interests include nonlinear filtering and reinforcement learning. He can be contacted at Ifes Campus Serra, Rodovia ES-010-Km 6,5—Manguinhos, 29173-087-Serra—ES, Brazil.

José Jaime da Cruz received the B.S.E.E. degree in 1974 and the M.Sc. degree in 1981, both from the University of São Paulo (USP), Brazil, and the D.Sc. degree in 1988 from the National Institute for Space Research (INPE), São José dos Campos, Brazil. He was at the Institute for Technological Research from 1976 to 1986 and at INPE from 1986 to 1989. He has been at USP since 1989, and is currently an associate professor.